

Machine Learning Internship Program

5-Week Weekly Internship Tasks

This internship program is designed to provide practical, hands-on experience in machine learning, from data preparation and exploratory analysis to model development, optimization, and the completion of a real-world machine learning project.

Week 1

Python, Data Preparation & Machine Learning Fundamentals

Objective

Build a strong foundation in Python, data handling, data preprocessing, and core machine learning concepts.

Tasks

1. Review Python fundamentals for data science.
2. Learn the basics of NumPy and Pandas.
3. Load and inspect a real-world dataset.
4. Identify:
 - Missing values
 - Duplicate records
 - Inconsistent or incorrect data
5. Clean and preprocess the dataset.
6. Understand the following machine learning concepts:
 - Features and target variables
 - Training and testing datasets
 - Classification and regression
 - Supervised and unsupervised learning
7. Split the dataset into training and testing sets.
8. Document the complete data preparation process.

Tools

- Python
- Jupyter Notebook or Google Colab
- Pandas
- NumPy

Deliverables

- Cleaned dataset
- Jupyter Notebook or Google Colab notebook
- Data preprocessing documentation
- Short explanation of the machine learning problem
- GitHub repository or Google Drive submission

Week 2

Exploratory Data Analysis & Feature Engineering

Objective

Understand the dataset, identify important patterns, and prepare meaningful features for machine learning models.

Tasks

1. Perform Exploratory Data Analysis (EDA).
2. Calculate descriptive statistics.
3. Analyze numerical and categorical variables.
4. Create meaningful data visualizations.
5. Identify correlations and relationships between variables.
6. Detect potential outliers.
7. Encode categorical variables.
8. Scale numerical features where necessary.
9. Create new features from existing variables.
10. Select relevant features for machine learning.

Tools

- Python
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn

Deliverables

- EDA notebook
- Data visualizations
- Feature engineering documentation
- List of selected features
- Minimum of 5 data-driven insights
- Updated dataset ready for machine learning

Week 3

Supervised Machine Learning: Classification & Regression

Objective

Build, train, evaluate, and compare supervised machine learning models.

Classification

Build at least **two classification models** using appropriate datasets.

Possible models include:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier
- K-Nearest Neighbors

Evaluate classification models using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Regression

Build at least **two regression models** using an appropriate dataset.

Possible models include:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

Evaluate regression models using:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Compare the performance of the models and explain which model performs best based on the selected evaluation metrics.

Tools

- Python
- Pandas
- Scikit-learn
- Matplotlib

Deliverables

- Machine learning notebook
- Minimum of two trained models
- Classification and regression model results
- Model comparison table
- Evaluation visualizations
- Short model performance report
- GitHub repository or Google Drive submission

Week 4

Model Optimization, Cross-Validation & Hyperparameter Tuning

Objective

Improve model performance and understand how to build reliable and generalizable machine learning models.

Tasks

1. Understand overfitting and underfitting.
2. Establish a baseline model.
3. Apply cross-validation.
4. Perform hyperparameter tuning.
5. Use optimization techniques.
6. Compare default and optimized models.
7. Analyze feature importance where applicable.
8. Evaluate model generalization.
9. Select the best-performing model using appropriate evaluation metrics.
10. Document all optimization decisions and results.

Tools

- Python
- Pandas
- Scikit-learn
- Matplotlib

Deliverables

- Optimized machine learning model
- Hyperparameter tuning notebook
- Cross-validation results
- Before-and-after performance comparison
- Feature importance visualization
- Model evaluation report
- GitHub repository or Google Drive submission

Week 5

Final Machine Learning Capstone Project

Objective

Apply the knowledge gained throughout the internship to complete an end-to-end machine learning project based on a real-world problem.

Machine Learning Workflow

Problem Definition → Data Collection → Data Cleaning → Exploratory Data Analysis → Feature Engineering → Model Development → Model Evaluation → Optimization → Final Model → Business or Real World Interpretation

Tasks

1. Select a real-world machine learning problem.
2. Find or collect an appropriate dataset.
3. Define a clear problem statement.
4. Identify the features and target variable.
5. Clean and preprocess the dataset.
6. Perform exploratory data analysis.
7. Engineer and select relevant features.
8. Train multiple machine learning models.
9. Compare model performance.
10. Optimize the best-performing model.

11. Evaluate the final model using appropriate metrics.
12. Interpret the model results.
13. Identify the limitations of the model.
14. Suggest possible future improvements.
15. Develop practical business or real-world recommendations.
16. Prepare a short project presentation.
17. Upload the completed project to GitHub or another portfolio platform.

Final Deliverables

1. Dataset

- Raw dataset
- Cleaned and processed dataset

2. Code

- Complete Jupyter Notebook or Google Colab notebook
- Well-organized and reproducible machine learning workflow

3. Analysis

- Exploratory Data Analysis
- Data cleaning and preprocessing documentation
- Feature engineering
- Feature selection

4. Machine Learning Models

- Multiple trained models
- Model comparison
- Final optimized model

5. Model Evaluation

Include appropriate evaluation metrics, visualizations, and interpretation based on the type of machine learning problem.

6. Final Project Output

- Key findings
- Model interpretation
- Business or real-world insights
- Recommendations
- Model limitations
- Suggested future improvements

7. Presentation and Portfolio

- Final project report
- Presentation slides, where applicable
- Short project presentation or demonstration
- GitHub repository or portfolio link

5-Week Internship Roadmap

Week	Focus Area	Main Deliverable
Week 1	Python, Data Fundamentals Preparation & ML	Cleaned and prepared dataset
Week 2	EDA & Feature Engineering	Analysis, visualizations, and selected features
Week 3	Supervised Machine Learning	Classification and regression models
Week 4	Model Optimization & Evaluation	Optimized and validated model
Week 5	Final ML Capstone Project	Complete end-to-end machine learning project

Expected Internship Outcome

By the end of the five-week internship, participants should be able to:

- Understand and prepare real-world datasets for machine learning.
- Perform exploratory data analysis and feature engineering.
- Build classification and regression models.
- Evaluate and compare machine learning models.
- Apply cross-validation and hyperparameter tuning.
- Interpret model results and feature importance.
- Develop an end-to-end machine learning project.
- Present their work professionally through a GitHub repository or portfolio.